In an ever changing business and economic environment, it is necessary to have an idea about the probable future course of events. Analysis of relevant time series helps to achieve this, especially by facilitating future business forecasts. Such forecasts may serve as crucial inputs in deciding competitive strategies and planning growth initiatives.

7.1 DEFINITION

Time series refers to any group of statistical information collected at regular intervals of time. Time series analysis is used to detect the changes in patterns in these collected data.

7.1.1 Definition by Authors

According to Mooris Hamburg "A time series is a set of statistical observations arranged in chronological order".

Ya-Lun-Chou : "A time series may be defined as a collection of readings belonging to different time periods of some economic variable or composite of variables".

W.Z. Hirsch says "The main objective in analyzing time series is to understand, interpret and evaluate change in economic phenomena in the hope of more correctly anticipating the course of future events".

7.1.2 Uses of Time Series

- Time series is used to predict future values based on previously observed values.
- Time series analysis is used to identify the fluctuation in economics and business.
- It helps in the evaluation of current achievements.
- Time series is used in pattern recognition, signal processing, weather forecasting and earthquake prediction.

It can be said that time series analysis is a big tool in the hands of business executives to plan their sales, prices, policies and production.

7.2 COMPONENTS OF TIME SERIES

The factors that are responsible for bringing about changes in a time series are called the components of time series.

Components of Time Series

1. Secular trend
2. Seasonal variation
3. Cyclical variation
4. Irregular (random) variation

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